



APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY
Thiruvananthapuram, Kerala, INDIA

Exam Centre: AHALIA SCHOOL OF ENGINEERING AND TECHNOLOGY

B.Tech S5 (R, S) Exam Nov 2025 (2019 Scheme) (S5 Result)

***TBP -To Be Published Soon**

MECHANICAL ENGINEERING[Full Time] (Generated on 21/03/2026 08:42 PM)	
Course Code	Course
MET301	MECHANICS OF MACHINERY
MET303	THERMAL ENGINEERING
HUT300	INDUSTRIAL ECONOMICS AND FOREIGN TRADE
MCN301	DISASTER MANAGEMENT
MET307	MACHINE TOOLS AND METROLOGY
HUT310	MANAGEMENT FOR ENGINEERS
MET305	INDUSTRIAL AND SYSTEMS ENGINEERING
MEL331	MACHINE TOOLS LAB-II
MEL333	THERMAL ENGINEERING LAB - I

Register No	Course Code (Grade)
ATP19ME006	MET303(F)
ATP19ME011	MET301(F), MET303(F)
ATP19ME014	MET301(F)
ATP19ME015	MET301(P)
ATP19ME017	HUT300(Absent), MET301(F), MET303(F), MET305(F)
ATP20ME003	MET301(P), MET303(F)
ATP20ME004	MET303(F)

ATP20ME013	HUT300(D), MET301(C), MET303(P)
ATP20ME015	MET303(F)
ATP21ME001	MET303(F)
ATP21ME002	HUT300(Absent), HUT310(Absent), MCN301(F), MET301(F), MET303(F), MET307(F)
ATP22ME003	MET301(C), MET303(F)
ATP22ME005	HUT300(C+), MET301(C)
ATP22ME006	MET301(Absent)
ATP22ME009	HUT300(Absent), MET307(Absent)
ATP23ME001	HUT300(F), MCN301(P), MEL333(A), MET301(F), MET303(F), MET305(P), MET307(F), MEL331(A)
ATP23ME002	HUT300(C+), MCN301(B), MEL333(A+), MET301(C), MET303(D), MET305(P), MET307(B), MEL331(S)
ATP23ME003	HUT300(C+), MCN301(C), MEL333(S), MET301(A), MET303(C+), MET305(B), MET307(B+), MEL331(S)
ATP23ME004	HUT300(D), MCN301(D), MEL333(A+), MET301(D), MET303(P), MET305(P), MET307(D), MEL331(A)
ATP23ME005	HUT300(C+), MCN301(D), MEL333(A), MET301(P), MET303(P), MET305(P), MET307(D), MEL331(S)
ATP23ME006	HUT300(P), MCN301(D), MEL333(C), MET301(F), MET303(F), MET305(P), MET307(F), MEL331(B+)
ATP23ME007	HUT300(F), MCN301(P), MEL333(A+), MET301(F), MET303(F), MET305(F), MET307(F), MEL331(S)
LATP22ME014	HUT300(Absent), MCN301(F)
LATP22ME016	MET303(Absent)
LATP23ME008	HUT300(C), MCN301(P), MEL333(S), MET301(C), MET303(D), MET305(B), MET307(C), MEL331(S)
LATP23ME009	HUT300(B), MCN301(B), MEL333(S), MET301(P), MET303(C), MET305(C), MET307(B), MEL331(S)
LATP23ME010	HUT300(B+), MCN301(D), MEL333(S), MET301(B), MET303(B), MET305(P), MET307(C+), MEL331(S)

CIVIL ENGINEERING[Full Time] (Generated on 21/03/2026 08:42 PM)

Course Code	Course
MCN301	DISASTER MANAGEMENT

CET305	GEOTECHNICAL ENGINEERING II
HUT300	INDUSTRIAL ECONOMICS AND FOREIGN TRADE
CET303	DESIGN OF CONCRETE STRUCTURES
CET307	HYDROLOGY AND WATER RESOURCES ENGINEERING
CET301	STRUCTURAL ANALYSIS I
CET309	CONSTRUCTION TECHNOLOGY AND MANAGEMENT
CEL331	MATERIAL TESTING LAB II
CEL333	GEOTECHNICAL ENGINEERING LAB

Register No	Course Code (Grade)
ATP19CE004	CET303(F), CET305(F), CET307(F)
ATP19CE022	CET303(F), CET305(F), CET307(F)
ATP20CE001	CET307(F)
ATP20CE034	CET301(F)
ATP21CE002	CET303(P), CET307(Absent)
ATP21CE007	HUT300(F), CET303(Absent), CET305(Absent), CET307(Absent)
ATP21CE014	CET301(P), CET303(P)
ATP21CE017	CET303(F), CET307(F)
ATP22CE006	CET301(Absent), CET307(Absent), CET309(Absent)
ATP22CE013	HUT300(P), MCN301(B)
ATP23CE001	MCN301(A+), CET301(D), CET303(D), CET305(D), CET307(C+), CEL331(S), CEL333(S), CET309(B)
ATP23CE003	MCN301(C+), CET301(P), CET303(D), CET305(P), CET307(P), CEL331(A+), CEL333(A), CET309(P)
ATP23CE004	MCN301(A+), CET301(C+), CET303(B), CET305(C+), CET307(B), CEL331(A+), CEL333(A+), CET309(C+)
ATP23CE005	MCN301(S), CET301(C), CET303(C+), CET305(B), CET307(B), CEL331(S), CEL333(A+), CET309(A)
ATP23CE006	MCN301(B+), CET301(F), CET303(F), CET305(P), CET307(F), CEL331(A+), CEL333(A), CET309(C)
ATP23CE007	MCN301(B+), CET301(C+), CET303(B), CET305(B+), CET307(A+), CEL331(S), CEL333(A+), CET309(B+)

ATP23CE008	MCN301(B+), CET301(F), CET303(F), CET305(P), CET307(F), CEL331(A), CEL333(A), CET309(D)
LATP21CE028	HUT300(F)
LATP22CE014	HUT300(F), MCN301(C+), CET303(P), CET305(P), CET307(F)
LATP22CE018	MCN301(C), CET305(P), CET307(F)
LATP22CE019	HUT300(F), MCN301(C+), CET303(F), CET305(F)
LATP23CE010	MCN301(B+), CET301(P), CET303(B), CET305(C+), CET307(B), CEL331(S), CEL333(A+), CET309(A+)
LATP23CE011	MCN301(A), CET301(D), CET303(P), CET305(C+), CET307(B), CEL331(S), CEL333(A), CET309(B+)
LATP23CE012	MCN301(B+), CET301(P), CET303(P), CET305(D), CET307(P), CEL331(A+), CEL333(S), CET309(A)
LATP23CE013	MCN301(P), CET301(F), CET303(F), CET305(F), CET307(F), CEL331(A), CEL333(A), CET309(F)

ELECTRONICS & COMMUNICATION ENGG[Full Time] (Generated on 21/03/2026 08:42 PM)	
Course Code	Course
ECT301	LINEAR INTEGRATED CIRCUITS
ECT303	DIGITAL SIGNAL PROCESSING
ECT305	ANALOG AND DIGITAL COMMUNICATION
ECT307	CONTROL SYSTEMS
HUT300	INDUSTRIAL ECONOMICS AND FOREIGN TRADE
HUT310	MANAGEMENT FOR ENGINEERS
ECL331	ANALOG INTEGRATED CIRCUITS AND SIMULATION LAB
ECL333	DIGITAL SIGNAL PROCESSING LAB
MCN301	DISASTER MANAGEMENT

Register No	Course Code (Grade)
ATP20EC001	HUT310(C), ECT301(F), ECT303(F), ECT305(F), ECT307(F)
ATP20EC005	ECT305(F), ECT307(F)
ATP20EC011	ECT307(Absent)
ATP20EC019	ECT307(A)

ATP20EC024	HUT300(Absent), ECT301(Absent), ECT305(F), ECT307(F)
ATP21EC001	ECT301(F), ECT303(F), ECT305(Absent), ECT307(F)
ATP21EC005	ECT301(F), ECT305(F), ECT307(F)
ATP21EC006	HUT300(F), ECT301(F), ECT303(F), ECT305(F), ECT307(F)
ATP21EC007	ECT305(D), ECT307(F)
ATP21EC008	ECT301(F), ECT305(F), ECT307(F)
ATP21EC012	ECT301(F), ECT307(F)
ATP21EC018	ECT301(F), ECT305(F), ECT307(F)
ATP21EC025	ECL331(D)
ATP21EC026	ECT305(P)
ATP21EC027	ECT305(D), ECT307(F)
ATP21EC041	ECT303(F), ECT307(F)
ATP22EC002	ECT305(F), ECT307(F)
ATP22EC003	ECT301(F), ECT307(P)
ATP22EC007	ECT305(F), ECT307(F)
ATP22EC010	ECT301(Absent), ECT305(Absent), ECT307(Absent)
ATP22EC012	ECT305(F)
ATP22EC013	ECT301(F), ECT305(F), ECT307(F)
ATP22EC015	ECT305(F), ECT307(P)
ATP22EC017	ECT305(P)
ATP22EC018	HUT300(P), HUT310(Absent), ECL331(F)
ATP22EC019	HUT300(P), ECT307(Absent)
ATP22EC020	ECT307(Absent)
ATP22EC021	HUT310(F), MCN301(B)
ATP22EC022	ECL331(F), ECL333(C)
ATP22EC023	ECT301(Absent), ECT303(Absent), ECT305(Absent), ECT307(Absent), ECL331(FE), ECL333(F)
ATP22EC027	HUT310(Absent), ECT301(Absent), ECT307(Absent)
ATP22EC028	HUT310(F), ECT303(Absent), ECT305(F), ECL331(F)
ATP22EC029	HUT310(Absent), ECT301(F), ECT305(Absent), ECT307(F), ECL331(B), ECL333(D)
ATP22EC031	HUT300(C), MCN301(A), ECT301(F), ECT303(F), ECT305(F), ECT307(F), ECL331(F), ECL333(B)

ATP22EC033	ECT301(P), ECT305(F), ECT307(Absent), ECL331(C+)
ATP22EC034	ECT307(Absent)
ATP22EC035	ECT307(D)
ATP22EC037	ECT307(P)
ATP22EC040	HUT300(C+), ECT301(F), ECT303(P), ECT305(Absent), ECT307(Absent), ECL331(F), ECL333(C)
ATP22EC041	ECT305(F), ECT307(F)
ATP23EC001	HUT300(C+), MCN301(A), ECT301(P), ECT303(P), ECT305(D), ECT307(P), ECL331(A), ECL333(B)
ATP23EC002	HUT300(P), MCN301(B), ECT301(P), ECT303(P), ECT305(P), ECT307(P), ECL331(A+), ECL333(A)
ATP23EC003	HUT300(C+), MCN301(A), ECT301(C+), ECT303(C), ECT305(B), ECT307(B), ECL331(A+), ECL333(S)
ATP23EC004	HUT300(C), MCN301(A), ECT301(A), ECT303(C+), ECT305(B+), ECT307(B+), ECL331(B+), ECL333(A)
ATP23EC005	HUT300(P), MCN301(B), ECT301(B), ECT303(C), ECT305(C), ECT307(D), ECL331(D), ECL333(A)
ATP23EC006	HUT300(D), MCN301(A), ECT301(B), ECT303(C+), ECT305(A), ECT307(B+), ECL331(A+), ECL333(S)
ATP23EC007	HUT300(C), MCN301(A), ECT301(B), ECT303(C), ECT305(C+), ECT307(B), ECL331(A), ECL333(S)
ATP23EC008	HUT300(D), MCN301(P), ECT301(F), ECT303(P), ECT305(F), ECT307(F), ECL331(F), ECL333(B)
ATP23EC009	HUT300(F), MCN301(P), ECT301(F), ECT303(F), ECT305(P), ECT307(F), ECL331(C), ECL333(B+)
ATP23EC010	HUT300(A), MCN301(A+), ECT301(B+), ECT303(B), ECT305(B+), ECT307(C), ECL331(S), ECL333(S)
ATP23EC011	HUT300(C), MCN301(A), ECT301(F), ECT303(P), ECT305(F), ECT307(F), ECL331(C+), ECL333(B+)
ATP23EC012	HUT300(P), MCN301(D), ECT301(B), ECT303(B), ECT305(B+), ECT307(C), ECL331(A+), ECL333(S)
ATP23EC013	HUT300(C), MCN301(B+), ECT301(C), ECT303(D), ECT305(D), ECT307(P), ECL331(B), ECL333(A+)
ATP23EC014	HUT300(B), MCN301(S), ECT301(D), ECT303(C), ECT305(B), ECT307(C), ECL331(C), ECL333(S)
ATP23EC015	HUT300(D), MCN301(C+), ECT301(F), ECT303(F), ECT305(F), ECT307(F), ECL331(D), ECL333(A)

ATP23EC016	HUT300(C), MCN301(A), ECT301(C), ECT303(P), ECT305(D), ECT307(P), ECL331(B), ECL333(A)
ATP23EC018	HUT300(F), MCN301(P), ECT301(P), ECT303(P), ECT305(F), ECT307(F), ECL331(A+), ECL333(A)
ATP23EC019	HUT300(D), MCN301(A), ECT301(B+), ECT303(D), ECT305(C+), ECT307(D), ECL331(A+), ECL333(S)
ATP23EC020	HUT300(A), MCN301(S), ECT301(A), ECT303(B+), ECT305(B+), ECT307(B+), ECL331(S), ECL333(S)
ATP23EC022	HUT300(F), MCN301(B+), ECT301(F), ECT303(P), ECT305(P), ECT307(F), ECL331(B+), ECL333(A)
ATP23EC023	HUT300(B), MCN301(C), ECT301(B+), ECT303(C), ECT305(C+), ECT307(D), ECL331(F), ECL333(C+)
ATP23EC024	HUT300(C+), MCN301(B+), ECT301(A+), ECT303(B), ECT305(B), ECT307(A), ECL331(A+), ECL333(A+)
ATP23EC025	HUT300(C+), MCN301(A+), ECT301(C+), ECT303(C+), ECT305(C+), ECT307(C), ECL331(D), ECL333(S)
ATP23EC026	HUT300(D), MCN301(C), ECT301(C+), ECT303(P), ECT305(D), ECT307(P), ECL331(F), ECL333(C+)
ATP23EC027	HUT300(C), MCN301(C), ECT301(D), ECT303(D), ECT305(D), ECT307(C+), ECL331(A+), ECL333(S)
ATP23EC028	HUT300(B+), MCN301(S), ECT301(A), ECT303(B), ECT305(A), ECT307(S), ECL331(S), ECL333(S)
ATP23EC029	HUT300(F), MCN301(A+), ECT301(D), ECT303(F), ECT305(C), ECT307(C), ECL331(D), ECL333(C+)
ATP23EC030	HUT300(B), MCN301(A), ECT301(B+), ECT303(B+), ECT305(B+), ECT307(C), ECL331(S), ECL333(S)
ATP23EC031	HUT300(D), MCN301(P), ECT301(F), ECT303(F), ECT305(F), ECT307(F), ECL331(F), ECL333(B+)
ATP23EC032	HUT300(D), MCN301(P), ECT301(F), ECT303(F), ECT305(F), ECT307(F), ECL331(A), ECL333(B)
ATP23EC033	HUT300(B+), MCN301(B), ECT301(B), ECT303(C+), ECT305(B+), ECT307(C), ECL331(A+), ECL333(S)
ATP23EC034	HUT300(A), MCN301(B), ECT301(A), ECT303(B), ECT305(S), ECT307(A+), ECL331(S), ECL333(S)
ATP23EC035	HUT300(C), MCN301(C), ECT301(A), ECT303(B), ECT305(A), ECT307(A), ECL331(S), ECL333(S)

ATP23EC036	HUT300(P), MCN301(P), ECT301(C+), ECT303(P), ECT305(P), ECT307(C), ECL331(C), ECL333(B+)
ATP23EC038	HUT300(F), MCN301(P), ECT301(F), ECT303(F), ECT305(F), ECT307(F), ECL331(D), ECL333(B)
ATP23EC039	HUT300(P), MCN301(C+), ECT301(B), ECT303(P), ECT305(C), ECT307(B), ECL331(A+), ECL333(A)
ATP23EC040	HUT300(B), MCN301(F), ECT301(P), ECT303(P), ECT305(C), ECT307(C), ECL331(F), ECL333(A)
ATP23EC041	HUT300(F), MCN301(F), ECT301(F), ECT303(F), ECT305(F), ECT307(F), ECL331(C), ECL333(C+)
ATP23EC042	HUT300(C), MCN301(P), ECT301(C), ECT303(D), ECT305(C), ECT307(C+), ECL331(B), ECL333(A)
ATP23EC044	HUT300(F), MCN301(P), ECT301(F), ECT303(F), ECT305(F), ECT307(F), ECL331(F), ECL333(B)
ATP23EC045	HUT300(C), MCN301(P), ECT301(P), ECT303(D), ECT305(P), ECT307(C), ECL331(F), ECL333(C+)
ATP23EC046	HUT300(C), MCN301(P), ECT301(F), ECT303(F), ECT305(F), ECT307(F), ECL331(C), ECL333(C+)
ATP23EC047	HUT300(F), MCN301(F), ECT301(F), ECT303(F), ECT305(F), ECT307(F), ECL331(D), ECL333(C+)
ATP23EC048	HUT300(A), MCN301(C), ECT301(B), ECT303(C), ECT305(D), ECT307(P), ECL331(A+), ECL333(A+)
ATP23EC049	HUT300(C+), MCN301(C), ECT301(D), ECT303(D), ECT305(P), ECT307(D), ECL331(A), ECL333(B+)
ATP23EC050	HUT300(B), MCN301(B+), ECT301(A), ECT303(C+), ECT305(B+), ECT307(B), ECL331(A), ECL333(S)
ATP23EC051	HUT300(P), MCN301(P), ECT301(F), ECT303(F), ECT305(F), ECT307(F), ECL331(F), ECL333(A)
ATP23EC052	HUT300(P), MCN301(C), ECT301(P), ECT303(P), ECT305(D), ECT307(D), ECL331(B), ECL333(B)
ATP23EC053	HUT300(F), MCN301(P), ECT301(F), ECT303(F), ECT305(F), ECT307(F), ECL331(A), ECL333(C+)
ATP23EC054	HUT300(C+), MCN301(D), ECT301(F), ECT303(P), ECT305(P), ECT307(F), ECL331(D), ECL333(C+)

Course Code	Course
EET301	POWER SYSTEMS I
EET303	MICROPROCESSORS AND MICROCONTROLLERS
EET307	SYNCHRONOUS AND INDUCTION MACHINES
EET305	SIGNALS AND SYSTEMS
HUT300	INDUSTRIAL ECONOMICS AND FOREIGN TRADE
HUT310	MANAGEMENT FOR ENGINEERS
EEL331	MICROPROCESSORS AND MICROCONTROLLERS LAB
MCN301	DISASTER MANAGEMENT
EEL333	ELECTRICAL MACHINES LAB II

Register No	Course Code (Grade)
ATP19EE001	EET305(Absent)
ATP20EE012	EET301(F), EET303(Absent), EET307(Absent)
ATP20EE013	EET307(Absent)
ATP21EE002	EET305(Absent)
ATP21EE006	HUT310(F), EET301(F), EET303(F), EET307(F)
ATP21EE009	EET303(Absent), EET305(Absent), EET307(Absent)
ATP21EE019	HUT310(F), MCN301(F), EET301(F), EET303(F), EET307(F)
ATP22EE003	HUT300(F), EET301(F), EET303(F), EET305(F), EET307(F)
ATP22EE005	EET301(F)
ATP22EE006	HUT300(F), EET301(F), EET305(Absent), EET307(F)
ATP22EE007	EET303(C), EET305(F), EET307(D)
ATP22EE013	EET307(Absent)
ATP22EE014	EET303(F), EET305(Absent), EET307(F)
ATP23EE001	HUT300(D), MCN301(A), EET301(F), EET303(P), EET305(F), EET307(P), EEL331(S), EEL333(A+)
ATP23EE002	HUT300(P), MCN301(C+), EET301(C), EET303(C), EET305(C), EET307(C), EEL331(F), EEL333(A+)
ATP23EE003	HUT300(F), MCN301(F), EET301(F), EET303(F), EET305(F), EET307(F), EEL331(P), EEL333(F)
ATP23EE004	HUT300(F), MCN301(P), EET301(F), EET303(F), EET305(F), EET307(F), EEL331(D), EEL333(C+)

ATP23EE006	HUT300(F), MCN301(D), EET301(D), EET303(F), EET305(F), EET307(D), EEL331(S), EEL333(B+)
ATP23EE008	HUT300(F), MCN301(P), EET301(F), EET303(F), EET305(F), EET307(F), EEL331(B+), EEL333(B+)
ATP23EE009	HUT300(F), MCN301(F), EET301(F), EET303(F), EET305(F), EET307(F), EEL331(C+), EEL333(C)
ATP23EE010	HUT300(F), MCN301(P), EET301(F), EET303(F), EET305(F), EET307(P), EEL331(A+), EEL333(B+)
ATP23EE012	HUT300(F), MCN301(P), EET301(C), EET303(P), EET305(F), EET307(F), EEL331(S), EEL333(A)
ATP23EE013	HUT300(F), MCN301(P), EET301(F), EET303(F), EET305(F), EET307(F), EEL331(B), EEL333(B+)
ATP23EE014	HUT300(F), MCN301(P), EET301(F), EET303(F), EET305(F), EET307(P), EEL331(B+), EEL333(A)
ATP23EE015	HUT300(F), MCN301(C), EET301(F), EET303(P), EET305(F), EET307(F), EEL331(A+), EEL333(A)
ATP23EE016	HUT300(C), MCN301(C+), EET301(F), EET303(F), EET305(F), EET307(P), EEL331(A+), EEL333(A)
ATP23EE017	HUT300(F), MCN301(A), EET301(D), EET303(F), EET305(P), EET307(D), EEL331(S), EEL333(S)
LATP20EE016	EET305(Withheld), EET307(Withheld)
LATP21EE021	EET305(F), EET307(F)
LATP22EE017	EET301(P), EET303(P)
LATP22EE018	HUT300(P), HUT310(Absent), EET303(P), EEL331(A+)
LATP23EE018	HUT300(F), MCN301(C+), EET301(P), EET303(F), EET305(F), EET307(F), EEL331(A), EEL333(A)
LATP23EE019	HUT300(F), MCN301(P), EET301(F), EET303(F), EET305(F), EET307(F), EEL331(A+), EEL333(B+)
LATP23EE020	HUT300(P), MCN301(C+), EET301(P), EET303(D), EET305(F), EET307(D), EEL331(A+), EEL333(S)
LATP23EE021	HUT300(P), MCN301(D), EET301(P), EET303(F), EET305(F), EET307(D), EEL331(A), EEL333(A)
LATP23EE022	HUT300(F), MCN301(B), EET301(P), EET303(F), EET305(F), EET307(F), EEL331(A+), EEL333(B+)
LATP23EE023	HUT300(D), MCN301(A), EET301(P), EET303(C+), EET305(P), EET307(D), EEL331(S), EEL333(S)

COMPUTER SCIENCE & ENGINEERING[Full Time] (Generated on 21/03/2026 08:42 PM)	
Course Code	Course
CST301	FORMAL LANGUAGES AND AUTOMATA THEORY
CST305	SYSTEM SOFTWARE
CST307	MICROPROCESSORS AND MICROCONTROLLERS
CST303	COMPUTER NETWORKS
HUT300	INDUSTRIAL ECONOMICS AND FOREIGN TRADE
CSL331	SYSTEM SOFTWARE AND MICROPROCESSORS LAB
CST309	MANAGEMENT OF SOFTWARE SYSTEMS
MCN301	DISASTER MANAGEMENT
CSL333	DATABASE MANAGEMENT SYSTEMS LAB

Register No	Course Code (Grade)
ATP19CS030	CST301(F), CST303(Absent), CST307(Absent)
ATP19CS036	CST301(C)
ATP19CS059	CST305(Absent), CST307(F)
ATP20CS035	CST307(F)
ATP20CS037	CST301(F), CST307(P)
ATP20CS054	CST301(F), CST303(F), CST305(F), CST307(F), CST309(C+), CSL331(C), CSL333(C+)
ATP21CS005	CST301(D)
ATP21CS008	HUT300(Absent), CST301(Absent), CST305(Absent), CST309(P)
ATP21CS013	HUT300(F), CST301(Absent), CST307(Absent)
ATP21CS016	HUT300(Absent)
ATP21CS018	HUT300(F), CST301(F), CST307(P)
ATP21CS020	HUT300(C), CST305(Absent), CST307(P)
ATP21CS024	CST307(D)
ATP21CS026	HUT300(Absent), CST301(Absent), CST307(Absent)
ATP21CS030	CST301(D), CST305(P), CST307(P)
ATP21CS031	HUT300(P), CST305(F), CST307(F)
ATP21CS042	CST301(F)

ATP21CS043	HUT300(C), CST301(F)
ATP22CS001	HUT300(Absent), CST301(F), CST303(F), CST307(F), CST309(Absent)
ATP22CS002	CST309(P)
ATP22CS005	HUT300(FE), CST305(Absent)
ATP22CS007	CST301(F), CST307(F)
ATP22CS010	CST307(C)
ATP22CS011	CST303(F)
ATP22CS014	MCN301(P), CST301(Absent), CST305(F), CSL331(C)
ATP22CS023	CST307(Absent)
ATP22CS026	CST301(Absent), CST305(Absent)
ATP22CS039	CST305(Absent), CST307(F)
ATP22CS040	HUT300(Absent), MCN301(B), CST307(F)
ATP22CS043	HUT300(Absent), MCN301(B), CST303(F), CST307(Absent)
ATP22CS050	HUT300(Absent), CST301(F), CST307(F), CST309(F)
ATP22CS057	CST307(Absent)
ATP22CS058	HUT300(FE), CST305(Absent)
ATP22CS059	CST301(D)
ATP22CS064	HUT300(Absent), MCN301(C), CST301(F), CST303(F), CST307(F)
ATP23CS001	MCN301(D), CST301(P), CST303(P), CST305(D), CST307(C), CST309(P), CSL331(A+), CSL333(S)
ATP23CS002	MCN301(D), CST301(F), CST303(F), CST305(F), CST307(F), CST309(F), CSL331(F), CSL333(B+)
ATP23CS003	MCN301(B+), CST301(P), CST303(P), CST305(C), CST307(C), CST309(C), CSL331(A), CSL333(A)
ATP23CS004	MCN301(A), CST301(P), CST303(C), CST305(C), CST307(B), CST309(B+), CSL331(A+), CSL333(A+)
ATP23CS006	MCN301(C), CST301(F), CST303(F), CST305(P), CST307(C), CST309(D), CSL331(B+), CSL333(B+)
ATP23CS007	MCN301(C+), CST301(F), CST303(F), CST305(P), CST307(D), CST309(D), CSL331(B+), CSL333(S)
ATP23CS008	MCN301(S), CST301(P), CST303(P), CST305(C+), CST307(C+), CST309(A), CSL331(A+), CSL333(A)

ATP23CS009	MCN301(C+), CST301(F), CST303(F), CST305(P), CST307(F), CST309(F), CSL331(C+), CSL333(B)
ATP23CS011	MCN301(P), CST301(F), CST303(F), CST305(F), CST307(F), CST309(F), CSL331(B), CSL333(B+)
ATP23CS012	MCN301(D), CST301(D), CST303(P), CST305(P), CST307(P), CST309(D), CSL331(A+), CSL333(B+)
ATP23CS013	MCN301(A+), CST301(D), CST303(A), CST305(C+), CST307(C), CST309(C), CSL331(A+), CSL333(S)
ATP23CS014	MCN301(S), CST301(F), CST303(F), CST305(C), CST307(C+), CST309(P), CSL331(C+), CSL333(B+)
ATP23CS015	MCN301(B+), CST301(P), CST303(P), CST305(P), CST307(D), CST309(C+), CSL331(B+), CSL333(A)
ATP23CS016	MCN301(A+), CST301(P), CST303(P), CST305(D), CST307(D), CST309(D), CSL331(C+), CSL333(A)
ATP23CS017	MCN301(B), CST301(P), CST303(C), CST305(C+), CST307(C), CST309(B+), CSL331(C+), CSL333(B+)
ATP23CS018	MCN301(A), CST301(P), CST303(P), CST305(D), CST307(C), CST309(D), CSL331(S), CSL333(S)
ATP23CS020	MCN301(A+), CST301(D), CST303(B), CST305(C+), CST307(Absent), CST309(A), CSL331(S), CSL333(A)
ATP23CS021	MCN301(Absent), CST301(F), CST303(F), CST305(F), CST307(F), CST309(F), CSL331(C+), CSL333(B)
ATP23CS022	MCN301(P), CST301(F), CST303(P), CST305(F), CST307(P), CST309(C+), CSL331(B+), CSL333(A)
ATP23CS023	MCN301(C), CST301(P), CST303(P), CST305(P), CST307(P), CST309(D), CSL331(C+), CSL333(B+)
ATP23CS024	MCN301(A+), CST301(B), CST303(B), CST305(C+), CST307(B+), CST309(B), CSL331(S), CSL333(S)
ATP23CS025	MCN301(A), CST301(P), CST303(P), CST305(B), CST307(C), CST309(C+), CSL331(A+), CSL333(A+)
ATP23CS026	MCN301(A+), CST301(P), CST303(P), CST305(C), CST307(D), CST309(C), CSL331(A+), CSL333(A+)
ATP23CS027	MCN301(P), CST301(F), CST303(F), CST305(F), CST307(F), CST309(P), CSL331(C+), CSL333(A+)
ATP23CS028	MCN301(P), CST301(F), CST303(F), CST305(F), CST307(F), CST309(F), CSL331(B+), CSL333(B)

ATP23CS029	MCN301(B+), CST301(B), CST303(B), CST305(C+), CST307(C+), CST309(C+), CSL331(A), CSL333(A)
ATP23CS030	MCN301(C+), CST301(D), CST303(B), CST305(C), CST307(C), CST309(C+), CSL331(A+), CSL333(S)
ATP23CS031	MCN301(C), CST301(F), CST303(F), CST305(F), CST307(P), CST309(D), CSL331(B), CSL333(A)
ATP23CS032	MCN301(C+), CST301(C), CST303(D), CST305(C), CST307(C), CST309(D), CSL331(S), CSL333(S)
ATP23CS033	MCN301(B), CST301(D), CST303(C+), CST305(B), CST307(C), CST309(P), CSL331(S), CSL333(A+)
ATP23CS034	MCN301(C), CST301(C+), CST303(B), CST305(B), CST307(C+), CST309(B), CSL331(S), CSL333(S)
ATP23CS035	MCN301(C+), CST301(C+), CST303(C), CST305(C+), CST307(C), CST309(C+), CSL331(S), CSL333(S)
ATP23CS036	MCN301(B+), CST301(C+), CST303(C+), CST305(B+), CST307(B+), CST309(B), CSL331(A+), CSL333(S)
ATP23CS037	MCN301(A), CST301(B+), CST303(B), CST305(A+), CST307(D), CST309(B), CSL331(S), CSL333(A+)
ATP23CS038	MCN301(F), CST301(F), CST303(F), CST305(P), CST307(F), CST309(F), CSL331(B+), CSL333(A)
ATP23CS039	MCN301(B), CST301(B), CST303(C), CST305(B), CST307(C), CST309(C+), CSL331(S), CSL333(A+)
ATP23CS040	MCN301(F), CST301(F), CST303(F), CST305(P), CST307(F), CST309(F), CSL331(B+), CSL333(B+)
ATP23CS041	MCN301(B), CST301(A), CST303(C), CST305(B+), CST307(C), CST309(P), CSL331(S), CSL333(S)
ATP23CS042	MCN301(P), CST301(P), CST303(F), CST305(F), CST307(F), CST309(F), CSL331(C), CSL333(B)
ATP23CS043	MCN301(C), CST301(C), CST303(P), CST305(B), CST307(D), CST309(B), CSL331(A+), CSL333(A+)
ATP23CS044	MCN301(B+), CST301(P), CST303(P), CST305(C+), CST307(B), CST309(D), CSL331(S), CSL333(A+)
ATP23CS045	MCN301(F), CST301(F), CST303(F), CST305(F), CST307(F), CST309(D), CSL331(S), CSL333(A+)
ATP23CS047	MCN301(B+), CST301(B), CST303(C+), CST305(A), CST307(A), CST309(B+), CSL331(S), CSL333(S)

ATP23CS048	MCN301(P), CST301(F), CST303(P), CST305(C), CST307(P), CST309(C), CSL331(C+), CSL333(B+)
ATP23CS049	MCN301(D), CST301(B), CST303(B), CST305(B), CST307(C+), CST309(C+), CSL331(S), CSL333(S)
ATP23CS050	MCN301(P), CST301(F), CST303(P), CST305(F), CST307(F), CST309(F), CSL331(C+), CSL333(A)
ATP23CS051	MCN301(C+), CST301(B+), CST303(B), CST305(A), CST307(B+), CST309(C), CSL331(S), CSL333(S)
ATP23CS052	MCN301(B), CST301(B+), CST303(B+), CST305(B), CST307(A), CST309(C+), CSL331(S), CSL333(S)
ATP23CS053	MCN301(C+), CST301(P), CST303(F), CST305(F), CST307(F), CST309(D), CSL331(C+), CSL333(A)
ATP23CS054	MCN301(B), CST301(F), CST303(P), CST305(F), CST307(D), CST309(P), CSL331(A+), CSL333(B+)
ATP23CS055	MCN301(C), CST301(A+), CST303(B+), CST305(A), CST307(B), CST309(B), CSL331(S), CSL333(S)
ATP23CS057	MCN301(P), CST301(F), CST303(F), CST305(P), CST307(F), CST309(F), CSL331(C+), CSL333(A+)
ATP23CS058	MCN301(C), CST301(P), CST303(D), CST305(D), CST307(P), CST309(P), CSL331(B), CSL333(A)
ATP23CS059	MCN301(D), CST301(P), CST303(P), CST305(P), CST307(P), CST309(P), CSL331(B+), CSL333(A+)
ATP23CS060	MCN301(P), CST301(P), CST303(P), CST305(P), CST307(P), CST309(P), CSL331(S), CSL333(A)
ATP23CS061	MCN301(C), CST301(F), CST303(F), CST305(P), CST307(P), CST309(F), CSL331(B+), CSL333(B+)
ATP23CS062	MCN301(F), CST301(F), CST303(F), CST305(F), CST307(F), CST309(F), CSL331(B), CSL333(B+)
ATP23CS063	MCN301(B), CST301(B), CST303(D), CST305(D), CST307(B), CST309(C+), CSL331(S), CSL333(S)
ATP23CS064	MCN301(A), CST301(P), CST303(P), CST305(C), CST307(A), CST309(C+), CSL331(C+), CSL333(A+)
ATP23CS065	MCN301(D), CST301(FE), CST303(F), CST305(FE), CST307(F), CST309(FE), CSL331(FE), CSL333(FE)
ATP23CS066	MCN301(C+), CST301(P), CST303(P), CST305(P), CST307(D), CST309(P), CSL331(B), CSL333(A+)

ATP23CS067	MCN301(A), CST301(A), CST303(P), CST305(A+), CST307(A+), CST309(B), CSL331(S), CSL333(S)
ATP23CS068	MCN301(B), CST301(C+), CST303(B+), CST305(C+), CST307(B), CST309(D), CSL331(S), CSL333(A+)
ATP23CS069	MCN301(B), CST301(F), CST303(F), CST305(P), CST307(F), CST309(C), CSL331(C+), CSL333(A+)
ATP23CS070	MCN301(C+), CST301(P), CST303(C), CST305(B), CST307(B), CST309(C+), CSL331(A), CSL333(A+)
ATP23CS071	MCN301(S), CST301(A+), CST303(A+), CST305(A+), CST307(A+), CST309(A), CSL331(S), CSL333(S)
ATP23CS072	MCN301(P), CST301(P), CST303(Absent), CST305(F), CST307(F), CST309(F), CSL331(F), CSL333(C+)
ATP23CS073	MCN301(B), CST301(P), CST303(F), CST305(D), CST307(F), CST309(F), CSL331(B+), CSL333(A)
ATP23CS074	MCN301(A), CST301(F), CST303(F), CST305(F), CST307(P), CST309(P), CSL331(B+), CSL333(B)
ATP23CS075	MCN301(A), CST301(C+), CST303(B), CST305(C+), CST307(A), CST309(B+), CSL331(A+), CSL333(A+)
ATP23CS076	MCN301(D), CST301(F), CST303(P), CST305(F), CST307(P), CST309(F), CSL331(S), CSL333(B)
ATP23CS077	MCN301(A), CST301(S), CST303(A+), CST305(A), CST307(A), CST309(A), CSL331(S), CSL333(S)
ATP23CS078	MCN301(D), CST301(F), CST303(F), CST305(F), CST307(F), CST309(F), CSL331(A), CSL333(B)
ATP23CS079	MCN301(C+), CST301(D), CST303(P), CST305(P), CST307(C+), CST309(P), CSL331(A), CSL333(A)
ATP23CS080	MCN301(C), CST301(F), CST303(F), CST305(P), CST307(P), CST309(D), CSL331(B), CSL333(C+)
ATP23CS081	MCN301(B+), CST301(P), CST303(C), CST305(D), CST307(B), CST309(C+), CSL331(B+), CSL333(A)
ATP23CS082	MCN301(P), CST301(F), CST303(F), CST305(F), CST307(F), CST309(F), CSL331(C+), CSL333(B)
ATP23CS083	MCN301(B+), CST301(P), CST303(P), CST305(P), CST307(D), CST309(C+), CSL331(A), CSL333(A)
ATP23CS084	MCN301(B+), CST301(B+), CST303(C+), CST305(C+), CST307(B), CST309(C), CSL331(A+), CSL333(S)

ATP23CS085	MCN301(P), CST301(C), CST303(F), CST305(F), CST307(F), CST309(P), CSL331(B), CSL333(B+)
ATP23CS086	MCN301(B), CST301(A), CST303(C), CST305(B+), CST307(B), CST309(C), CSL331(S), CSL333(S)
ATP23CS087	MCN301(C), CST301(F), CST303(F), CST305(F), CST307(F), CST309(F), CSL331(B+), CSL333(A)
ATP23CS088	MCN301(P), CST301(F), CST303(F), CST305(F), CST307(F), CST309(F), CSL331(C+), CSL333(B)
ATP23CS089	MCN301(C+), CST301(P), CST303(F), CST305(F), CST307(F), CST309(F), CSL331(F), CSL333(A+)
ATP23CS090	MCN301(A), CST301(B), CST303(C+), CST305(B+), CST307(B), CST309(C), CSL331(S), CSL333(S)
ATP23CS091	MCN301(P), CST301(F), CST303(F), CST305(F), CST307(F), CST309(F), CSL331(B), CSL333(B+)
ATP23CS092	MCN301(C+), CST301(P), CST303(P), CST305(P), CST307(P), CST309(P), CSL331(C+), CSL333(A+)
ATP23CS093	MCN301(B), CST301(C), CST303(P), CST305(C), CST307(P), CST309(D), CSL331(S), CSL333(A+)
ATP23CS094	MCN301(B), CST301(P), CST303(C), CST305(P), CST307(P), CST309(C), CSL331(B+), CSL333(A)
ATP23CS095	MCN301(B+), CST301(P), CST303(B), CST305(B), CST307(C), CST309(C+), CSL331(B), CSL333(A+)
ATP23CS096	MCN301(C), CST301(F), CST303(P), CST305(C), CST307(P), CST309(D), CSL331(C+), CSL333(A)
ATP23CS098	MCN301(B+), CST301(P), CST303(P), CST305(C), CST307(B), CST309(C), CSL331(S), CSL333(B)
ATP23CS099	MCN301(F), CST301(F), CST303(F), CST305(F), CST307(F), CST309(F), CSL331(D), CSL333(B+)
ATP23CS100	MCN301(S), CST301(C+), CST303(B+), CST305(A), CST307(B), CST309(B+), CSL331(S), CSL333(A+)
FIT23CS006	MCN301(P), CST301(P), CST303(D), CST305(C), CST307(F), CST309(P), CSL331(A), CSL333(A+)
LATP22CS067	HUT300(Absent), CST301(Absent), CST305(Absent)
SIT21CS033	CST301(F), CST303(P), CST305(F), CST307(P)

